



Registers of beneficial owners based on blockchain technology: Implications for the accounting profession

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ABSTRACT

Central registers of beneficial owners are fraught with legal loopholes, and trust, privacy, and verification issues that devalue accountants' anti-money laundering compliance efforts. Yet, central registers have become important policy tools for governments in enhancing corporate transparency. We propose a blockchain-based solution that provides a more open, verifiable, and secure framework for registering beneficial ownership information, while ensuring greater transparency and trust. This paper examines the role of blockchain technology within registers of beneficial owners and highlights important implications for the accounting sector. We offer a fresh perspective into how blockchain technology supports company disclosure and contribute new insights into research on 'digital trust'. This paper bridges the gap between the conceptual and real-world implementation of registers of beneficial owners and promotes a more nuanced understanding on how new digital infrastructures impact accountants' compliance responsibilities.

1. Introduction

Various corporate vehicles, including companies, trusts, foundations, partnerships, and other legal entities, engage in a diverse range of commercial and entrepreneurial activities. While these entities play an essential and legitimate role in the global economy, their distinctive legal status also renders them susceptible to misuse through complex schemes designed to obscure the identities of their true beneficial owners (Financial Action Task Force [FATF], 2023a). A beneficial owner is defined as the natural person(s) who ultimately owns or controls assets or an interest in a company or other legal arrangement (FATF, 2023a). This may not be the same person listed as the legal owner. Moreover, individuals can conceal the identity of a company's beneficial owner in order to hide illicit wealth, launder money and evade tax. High profile exposés, like the Panama Papers and Paradise Papers,¹ have helped to uncover the various and dubious offshore dealings involving

the rich and famous, political elite, and regulated sector in abusing the financial system (Gilmour, 2020). Thus, beneficial ownership has evolved as a key policy concept in the global fight against financial crime (FATF, 2023a).

The Financial Action Task Force (FATF), functioning as an independent inter-governmental body, plays a crucial role in shaping policies aimed at safeguarding the global financial system from money laundering, terrorist financing, and the financing of weapons of mass destruction (Goldbarsht, 2017, p. 7). While FATF's recommendations are not legally binding, they are widely acknowledged as the international benchmark for anti-money laundering (AML) and counter-terrorist financing (CTF) standards (Goldbarsht, 2020). Impressively, these recommendations exhibit a high level of compliance across more than 180 member states.

The FATF's proactive approach to addressing money-laundering practices originated in 1991 when it began contemplating the role of

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¹ The Panama Papers constitute a vast collection of several million documents pilfered from Mossack Fonseca, a Panamanian law firm, and subsequently shared with investigative journalists worldwide. These papers unveiled a worldwide network of shell companies, established for various purposes, both legitimate and otherwise, by individuals across numerous countries. Notably, implicated individuals included former prime ministers in the United Kingdom and Iceland. Similarly, the Paradise Papers also encompass a substantial repository of data related to shell companies on a global scale. In general, the Paradise Papers revealed fewer instances of illegal arrangements compared to the Panama Papers, providing insights into the intricate web of global financial structures (see Littrell, 2024).

'non-traditional financial institutions or professions' that might be vulnerable to such activities (FATF, 1992). By 2001, the FATF recognised a mounting concern—that criminals were increasingly seeking assistance from professionals in laundering funds, acting as financial intermediaries, and offering expert advice. This phenomenon, known as the 'displacement effect', denotes the tendency of money launderers and terrorists to turn to professionals outside the financial sector due to heightened regulations within that industry (FATF, 2003). Notably, accountants have been officially recognised by the FATF as one of the seven gatekeeping sectors² in combating money laundering and terrorist financing (Goldbarsht and Benson, 2023).

Accountants, therefore, are crucial gatekeepers in upholding financial transparency and integrity. Accountants are 'obliged entities', mandated under relevant anti-money laundering (AML) and beneficial ownership disclosure regimes. They are responsible for assessing and understanding the money-laundering and terrorist-financing risk to which they are exposed and for implementing appropriate mitigation measures based on the level of risk (Shahaab et al., 2023). Accountants must, according to the FATF Recommendations 22 and 23, undertake customer due diligence (CDD), Know Your Customer (KYC) and record keeping, which involves supervising and monitoring transactions on behalf of their clients, identifying and keeping beneficial ownership records, and reporting discrepancies through official channels. Therefore, accountants have an important role in mitigating and managing money-laundering and terrorist-financing risk. Integrating innovative technologies, like blockchain, into these processes can empower accountants to more effectively fulfil their compliance roles.

Blockchain technology³ has become an important enabler for the regulated financial system, providing the digital infrastructure for secure and transparent transactions across various industries, revolutionising the way data is stored, accessed, and verified (Segura Ibarra et al., 2024; Xu and Viriyasitavat, 2019; Yessenbayev et al., 2024). A blockchain is a digital record of transactions across a peer-to-peer computer network and type of Distributed Ledger Technology (DLT) "for which transactions are grouped together and validated in blocks" (Rico-Peña et al., 2023, p. 2). Data is encrypted via the computer process of cryptography to provide the highest level of security to the blockchain environment, ensuring data is kept secure via permanent records (Khawaja and Javidroozi, 2023). Governments and public-sector bodies leverage blockchain technology to transition away from siloed and inefficient centralised systems. As such, blockchain technology operates to gain users' trust through a consensus protocol in which all users must agree before transactions can be validated.

This paper advances on Miraz and Ali (2018) notion that blockchain serves as an ideal 'trust machine'. Trust is an inherent feature of blockchain technology—the system requires no faith in an intermediary or other users to ensure data recorded on the digital ledger are validated and forever preserved (Miraz and Ali, 2018). However, we argue that in order to function as the ideal 'trust machine', blockchain must be fully transparent and accessible to stakeholders with tangible benefits to wider society. In this paper, we consider accountants (as obliged entities, like legal professionals and financial institutions) to be key stakeholders that should engage in and rely on the blockchain system to effectively comply with appropriate AML/CTF compliance rules. For accountants to fulfil their CDD and KYC obligations, they must have ready access to beneficial ownership information, including the digital

infrastructures to aid the use and exchange of information on their clients and associated companies.

We contribute to extant literature on accounting in several ways. First, we advance the growing debate on technological innovations that provide the regulated sector with 'digital trust' (see Park et al., 2018; Sahut et al., 2022, p. 2; Shin, 2019) by offering unique insights into the concept of adopting trusted and transparent beneficial ownership registers based on blockchain technology. In doing so, we recognise concerns around the lack of theory-based studies on blockchain applications (see Li et al., 2023; Lu, 2019) and contribute to addressing these concerns through the development of our novel theoretical model. Second, we examine the global efforts to enhance corporate transparency through the disclosure of beneficial ownership, including digital identifiers that can help track disclosure compliance (Leung et al., 2022; Sullivan and Burger, 2017; Sullivan and Tyson, 2023). Our paper is the first study of its kind to propose a theoretical framework that links the application of blockchain technology for registers of beneficial owners with greater regulatory oversight. We present a model that bridges the gap between technological innovation and corporate governance that, not only, enriches the academic discourse, but also, offers actionable solutions to pressing challenges in global corporate transparency.

Research has recognised the challenges in enhancing transparency of accounting information across various sectors of industry, from health care (Kuo et al., 2017; Robbins and Lapsley, 2015) and education (Angluin and Scapens, 2000) to banking (Ertan et al., 2017; Manganaris et al., 2017) and public-service organisations (Shahaab et al., 2023; Warren et al., 2015). Segura Ibarra et al. (2024) provide a comprehensive survey on how blockchain can address challenges, like data manipulation, intellectual property theft, and lack of transparency, while also promoting sustainability, collaboration, and trust across various stages of product development. Hamidi et al. (2024) propose a maturity model to guide practitioners in assessing their organisation's digital readiness and logistics capabilities, through integrating blockchain technologies to transform the maritime industry. Yet, little research has considered how digital technologies enhance the trust and transparency of company ownership records for accounting purposes (Dai and Vasarhelyi, 2017; Pimentel and Boulianne, 2020). Our research provides a fresh perspective into how blockchain technology can support company disclosure and explores the implications for the accounting profession.

This paper is structured as follows. First, it outlines the theoretical framework that underpins our work. The paper will then explore the role of beneficial ownership transparency in combating financial crime, examine the function of registers of beneficial owners, and address the associated challenges in implementing them. It will then identify the characteristics of blockchain technology and importance of Regulated Technology (RegTech) before moving onto exploring blockchain's application beyond cryptocurrency. The paper will address concerns about the need for enhanced transparency while safeguarding fundamental rights to privacy, before drawing conclusions.

2. Theoretical framework

2.1. Trust, transparency, and access

Blockchain technology offers an advantage over many centralised systems by eliminating the need for a controlling authority (Miraz and Ali, 2018). Trust is an inherent feature of blockchain technology, which arguably contributed to the early success of the cryptocurrency, Bitcoin (Marella et al., 2020; Simser, 2015). Albayati et al. (2020, p. 5) note that trust refers to the "level of comfort, confidence, and security that consumers have when using technologies." Trust suggests a reliable exchange of expectations in a manner that can be reasonably expected (Centobelli et al., 2022). Blockchain achieves trust through a reliable and shared mechanism that permanently records transactions (Centobelli et al., 2022). It might also help to overcome issues arising

² The other six sectors are casinos and other gambling businesses; dealers in real estate and high value items; company and trust service providers; notaries; legal professionals and auditors; and investment advisers (FATF, 2002, para. 87).

³ There exists different blockchain architectures (e.g., Hyperledger and Ethereum) with performance differences. A full discussion of these topics is beyond this paper's scope (see Capocasale et al., 2022; Guggenberger et al., 2022).

from a lack of transparency within domestic tax regimes (Delmotte, 2023; Kabir, 2021; Niu et al., 2022). As Kabir (2021) demonstrates, trust is an important factor in determining whether stakeholders intend to adopt blockchain technology for tax systems. Importantly, users of the blockchain system place trust in both the computer code, and the community that verifies the data using that code (Simser, 2015).

Yet, transparency and accessibility are just as important as trust for the effective functioning of blockchain-based solutions (Rane et al., 2021). As Gan and Lau (2024) argue, blockchain affords enhanced transparency of data in a manner that relates to the specific needs of the data user by being able to authenticate and share data. Transparency promotes the perceived usefulness of blockchain within society, referring to how effectively an institution informs the community about its functioning and practices (Baudet and Medina, 2023). Transparency also encompasses a degree of accountability (Baudet and Medina, 2023). Data on beneficial ownership must be fully transparent to hold professionals accountable for identifying corporate ownership and informing risk-based decisions in compliance with relevant AML/CTF regulations.

Research has highlighted the theory of ‘information asymmetry’, whereby inadequate sharing of information can lead to unequal access to that information (Acharya and Lambrecht, 2015; Clarkson et al., 2007; Rinaldo and Somogyi, 2021). This can hinder fair and equitable market competition in financial or corporate sectors, leading to unethical and morally haphazard decision-making processes (Kim and Kauffman, 2024). There has been little discourse of information asymmetry risks beyond finance, though emerging literature explores blockchain's role in mitigating them. Blockchain can reduce information asymmetry by making financial and contractual information more accessible and verifiable (Javaid et al., 2022). Chen (2023) argues, the decentralised, immutable and transparent nature of blockchain enables small and medium enterprises to access finance and openly share operational information to their banks or creditors. Jeanneret Medina et al. (2024) found adopting blockchain can minimise information asymmetry in supply-chain management relationships by avoiding opportunistic behaviours of agents.

Any blockchain-based system must be accessible and user-friendly to enable the robust disclosure of company ownership, aid the exchange of information, and foster a more inclusive and equitable ecosystem (El Koshiry et al., 2023). Accessibility of the system ensures that all relevant stakeholders are empowered to contribute to the collective aim of combating financial crime. As such, accessibility improves social integration and overall function of the system. We argue that only when trust is accompanied with transparency and open access can blockchain function as the ideal ‘trust machine’, produce meaningful outputs, and support effective AML/CTF compliance. Our new conceptual model, encompassing trust, transparency, and access within such a system, is represented in Fig. 1.

3. Beneficial ownership transparency

3.1. Enhancing beneficial ownership transparency

Beneficial ownership transparency, through the disclosure of company ownership, has become a fundamental requirement under international Anti-Money Laundering and Counter Terrorist Financing (AML/CTF) standards (Konovalova et al., 2023; Thomas-James, 2022). These standards govern the exchange of tax information, mandating transparency in two main ways: (1) through the provision of information upon specific requests, and (2) through the automatic exchange of financial account information. The dual approach ensures accessibility of essential data about the true owners of legal entities. Beneficial ownership recognises the fundamental separation of the legal owner from those benefiting or enjoying a company asset, helping to settle the rightful owner when determining trust arrangements, commercial acquisitions, and other business dealings (Gillis, 2019; Thorpe, 2021).

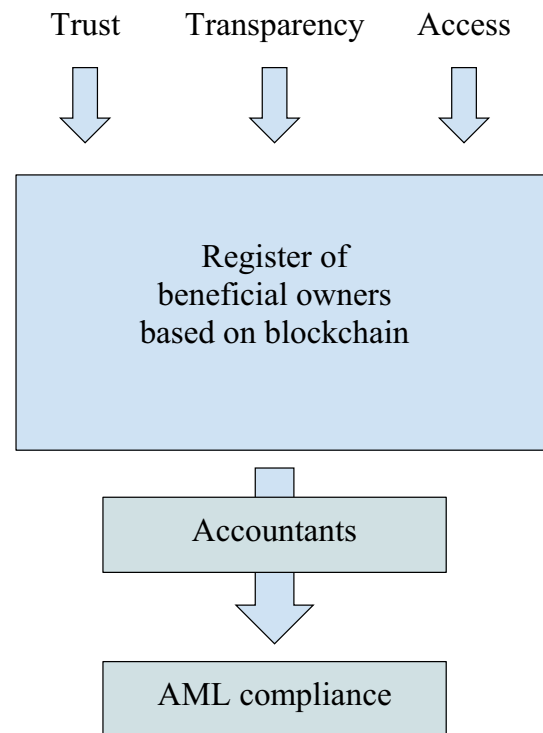


Fig. 1. A theoretical framework for a register of beneficial owners based on blockchain technology. For a blockchain to function as the ideal ‘trust machine’, the blockchain must be fully transparent and accessible for accountants to engage with reliably to support effective AML compliance.

Improved transparency is core to advancing corporate governance and developing sound and reliable policy recommendations (Goergen and Mira, 2023). It ensures market investors can make better informed decisions with greater confidence, while encouraging a culture of trust. When investors are aware of an entity's beneficial owners, they can better assess risk involved in investing and identify potential unethical corporate practices (Sihombing, 2023).

Transparency campaigners have understandably criticised strong tax avoidance schemes that benefit large multinational companies, the wealthy or political elite. When the wealthy or political elite exploit legal loopholes to minimise their tax obligations, it creates a perception of inequality and injustice. A lack of transparency in turn damages trust in democracy and the integrity of the wider financial system (Nauli et al., 2023). Aggressive tax avoidance by multinational corporations can result in governments losing tax revenue, which threatens spending on healthcare, education, and public infrastructure; thus, hindering governments' capacity to address pressing societal needs.

Although literature has helped to highlight the flaws inherent within international transparency initiatives (see Cobham, 2013; Gardner et al., 2019; Gillies and Heuty, 2011; Knobel, 2023) little study has centred on the implications of such policies on accountants (Konovalova et al., 2023). Accountants operate at the intersection of regulatory changes and corporate practices focused on enhancing transparency. Therefore, new research must seek to understand how these professionals navigate and respond to transparency initiatives based on innovative technologies, like blockchain. Building on the importance of transparency initiatives, the next section examines how public registers, as a key tool, can further enhance transparency and support the effective implementation of regulatory frameworks.

3.2. Public registers as tools for enhancing transparency

The status of beneficial ownership registries globally is diverse and at

various stages of progress. Nearly 100 jurisdictions have laws mandating beneficial ownership registration, but many contain gaps undermining their value (Knobel, 2023; Matras, 2023). The management of these registries varies, with the commercial register, a new beneficial ownership register, tax administration, financial intelligence units, and central banks being the standard authorities in charge. While public access to ownership information has expanded, the scope of the registration frameworks varies with limited types of companies (Knobel and Lorenzo, 2022).

For instance, the United Kingdom has implemented multiple registers, including the 'People with Significant Control' (PSC) register, which covers UK-based private companies (those limited by shares and by guarantee) limited liability partnerships (LLPs) and Societates Europaeae (SEs) and most public limited companies, among others (Companies House, 2023). However, the PSC register does not cover sole traders, overseas-registered companies operating within the UK, and limited partnerships (Companies House, 2023). Nonetheless, the PSC register is publicly available. The UK's register of trusts, however, is only available to those with a 'legitimate interest', e.g., law-enforcement and regulated bodies (Morgan, 2022). More recently, the UK's Register of Overseas Entities came into force through the Economic Crime (Transparency and Enforcement) Act 2022 and covers foreign entities owning property in the UK (Advani et al., 2023).

Elsewhere, the United States is developing transparency through the Corporate Transparency Act, which aims to address the issue of shell companies—entities that exist on paper but conduct little to no business (Azrilyant, 2021). Despite their legitimate uses (e.g., company mergers and acquisitions) the US Government Accountability Office (2019) recognises that anonymous shell companies can be used illegally as vehicles to hide beneficial ownership (see also, Azrilyant, 2021; Gillis, 2019). However, companies having over twenty full-time employees, public-listed companies, and many private entities and limited liability companies are exempt from disclosing their beneficial owners (Gilmour, 2022b).

Meanwhile, the European Union has introduced updated Anti-Money Laundering Directives focusing on measures to enhance beneficial ownership transparency (Gilmour, 2022a, 2022b). The FATF has also adopted changes to Recommendation 24 that aims to strengthen how international governments respond to the concealment of beneficial ownership of legal persons (FATF, 2023a). However, it remains to be seen what impact these changes will have on beneficial ownership transparency in the long term. Similarly, in India, the move towards beneficial ownership transparency is marked by recent legislative advancements, achieving compliance with the G20 Principle 1 (Transparency International, 2015). Here, companies are required to maintain beneficial ownership information and can now approach the National Company Law Tribunal to compel individuals to disclose beneficial ownership; although, accessibility improvements are pending (Gaur, 2023; Transparency International, 2015). Nonetheless, significant challenges in implementing registers of beneficial owners remain.

3.3. Challenges of implementing beneficial ownership registers

The global implementation of effective beneficial ownership frameworks has encountered substantial challenges. According to FATF (2022, p. 6) only 52 % of jurisdictions have successfully implemented laws to identify beneficial owners, suggesting a lack of political will or other procedural hurdles. Moreover, the application of beneficial ownership is problematic; the lack of a strictly defined natural person creates an open-ended interpretation, which complicates how beneficial ownership data is entered and verified in a register (Matras, 2023).

As definitions of beneficial ownership vary, some jurisdictions focus only on ownership, voting rights, rights to benefits, or combinations of these elements (Knobel and Lorenzo, 2022). The triggers for registration also differ, with local incorporation, tax residency, and local operations or assets being common conditions. Thresholds in definitions pose a

challenge, as they can allow for the concealment of beneficial owners. Thresholds set a minimum percentage of ownership, control, or benefits that an individual must meet to be recognised as a beneficial owner. A common threshold is 'more than 25%', the threshold suggested by the FATF (2023b) in its updated recommendations (Matras, 2023). Yet, this arbitrary limit creates a loophole whereby ownership can be spread across multiple individuals, each owning less than the threshold, allowing the concealment of beneficial ownership (Gilmour, 2020). Even with lower thresholds, there remains potential for obfuscation; although, they offer more transparency than higher thresholds (Gilmour, 2020; Knobel and Lorenzo, 2022). Nonetheless, difficulties arise with pooled investment vehicles located in non-FATF member jurisdictions, leading to gaps in identifying the beneficial owner when no individual holds a majority stake (Pai, 2023).

In response to these challenges, the global framework for tracing beneficial ownership has been evolving to accommodate the changing dynamics of the economy (Vaidyanathan et al., 2018). The trend of employing registers that enable the identification of beneficial owners is characterised by FATF guidance and various geopolitical events, which are further complicated through shell companies and complex ownership structures. The guidance on beneficial ownership transparency focuses on legal arrangements and clarity in definitions (FATF, 2023a).

Research has also highlighted a lack of accuracy and verification concerning beneficial ownership records held within central registers (Markle and Kiepe, 2022; Matras, 2023; Global Witness, 2018; Knobel, 2019; OpenOwnership and Global Witness, 2017). Ensuring the verification and accuracy of the information in a beneficial ownership registry is crucial to the efficacy of the system, otherwise this issue becomes the systems' weak link in the fight against financial crimes. Although centralisation has been highlighted to improve accessibility and consistency of information (Knobel and Lorenzo, 2022) adopting centralised systems alone may struggle to overcome the multitude of challenges facing the regulated accounting sector in enhancing beneficial ownership transparency.

4. Blockchain technology as a solution

The term 'blockchain' originates from the Bitcoin Protocol, outlined in the seminal whitepaper, 'Bitcoin: A Peer-to-Peer Electronic Cash System' authored by the mysterious Satoshi Nakamoto (Nakamoto, 2008). Nakamoto (2008) articulates the prerequisites for establishing an electronic cash system founded on an immutable digital accounting ledger, eliminating the need for trusted third-party intermediaries to determine truth or declare ownership. While the term 'blockchain' is notably absent in Nakamoto's (2008) whitepaper, the Bitcoin Protocol incorporates an economic incentive structure and cryptographic principles to form the bitcoin ledger. The protocol mandates the aggregation of bitcoin transactions into blocks and the linkage of valid blocks through a chain of cryptography, laying the foundation for what we now commonly refer to as blockchain (Nakamoto, 2008; Kyles, 2022).

Blockchain technology organises data on a peer-to-peer network of computers, securely recording data modifications, such as transactions, in time-stamped blocks. Blockchain works through cryptography, a highly secure computer process that involves solving computer-based mathematical calculations to convert data to a random string of text, or hash (Gilmour, 2022a). Making alterations to the blockchain is nearly impossible, as changing the data inputted in a block, or attempting to resubmit altered blocks to the chain would require a monumental amount of computing power (Gilmour, 2022a). These blocks are interconnected, forming a chain, and are confirmed and stored by all network users, making the ledger difficult to tamper with; therefore, enhancing trust (Segura Ibarra et al., 2024).

Blockchain has built-in rules for nodes to verify, and these rules do not need to be hard-coded (Casino et al., 2019). Each participating node must agree to the blockchain's unique protocol built into the algorithm when updating the blockchain, otherwise, the data will be rejected, and

a new verification process will commence (El Koshiry et al., 2023; Gilmour, 2022a). This consensus protocol process ensures proof of work that is displayed openly in a verifiable way to create a public ledger that cannot be refuted later. Key features include cryptographic digital signatures that ensure data authenticity, a distributed notary network for transaction verification, and a consensus algorithm for fault tolerance.

The most successful aspect is the smart contract, a digital agreement written in computer code and programmed to execute automatically once certain predefined conditions have been met (Gilmour, 2023; Hamidi et al., 2024). This allows users to define business logics as rules in the form of an automated programme to facilitate legal, financial, and social agreements, while assuring the terms of the agreement without a trusted third party (Gilmour, 2023; Khan et al., 2021). As Luo and Yu (2022) note, blockchain ventures leverage smart contracts for fund-raising investments. The use of smart contracts with blockchain platforms can automate processes like share conversions, thereby reducing administrative burdens and the potential for human error (Dentons, 2018). This enables the blockchain to serve as the foundation of various decentralised applications written as smart contracts.

4.1. Regulated technology

Integrating blockchain into beneficial ownership frameworks and existing tax and financial systems is complex. Financial institutions are integrating blockchain for trade digitisation and cross-border payments (Vaidyanathan et al., 2018). Thus, key area of growth concerns regulatory compliance and regulatory technology. The role of 'Regulated Technology' (RegTech) in overcoming the hurdles of cross-border data sharing and the harmonisation of regulations is becoming increasingly valued. RegTech refers to the application of innovative technologies aimed at enhancing the effectiveness and efficiency of meeting regulatory and compliance obligations. RegTech also presents a potential solution to the issue of de-risking (Woodsome and Ramachandran, 2018). The scope of RegTech extends to various regulatory compliance activities, encompassing areas like AML and CTF (Institute of International Finance, 2017). For Zetzsche et al. (2020, p. 202) blockchain acts as an automated form of 'embedded supervision' for which compliance, monitoring and supervision are implemented as inherent functions to fulfil compliance requirements.

Specifically, DLT can securely store and share 'Know Your Customer' (KYC) information and facilitate cheaper, more secure international payments (Woodsome and Ramachandran, 2018). Yet, effective compliance is costly. According to the British Banks' Association over £890 million is spent on financial crime compliance every year (de Long et al., 2017). It is also onerous and time consuming to perform enhanced diligence checks (for instance, when dealing with Politically Exposed Persons (PEPs) or other high-risk entities). The relevance of blockchain technology to beneficial ownership tracing lies in its secure and transparent nature, which can aid in the accurate identification of beneficial owners, thereby enhancing the transparency and integrity of financial transactions and compliance processes. KYC data is currently confined within individual organisational silos of financial institutions, creating a need for centralised intermediaries to improve market-wide efficiency (Institute of International Finance, 2017). According to Guo and Liang (2016, p. 8) blockchain can help financial institutions avoid unnecessary processes and save costs by automating encryption and sharing of customer information and transaction records between institutions. Still, as decentralised applications grow in popularity, there becomes a need to enhance the efficiency of the blockchain foundations. Various new typologies are emerging, each contributing to the improvement of blockchain technology in different ways (Rico-Peña et al., 2023). We now consider the applications of blockchain technology within public-data systems further in the following section.

4.2. The application of blockchain technology in public-data systems

Blockchain remains a relatively new technology for many sectors of society (Attaran, 2022; El Koshiry et al., 2023). Although, it is emerging in various sectors beyond cryptocurrency, generating a list of new concepts and applications with public data systems, such as Web3, Tokenization, and Distributed Autonomous Organizations (DAOs). Navadkar et al. (2018) discussed the potential applications of blockchain in the public sectors, including civil registration, healthcare, education, agriculture, public safety (see also, Garg and Alam, 2023; Ølnes et al., 2017). Here we provide a glimpse of the studies presenting concrete applications.

Blockchain is supporting the governance of public-sector systems through enhanced transparency and trust (Ølnes et al., 2017). The application of blockchain technology is relevant for government agencies in protecting trusted records and simplifying interactions with citizens (Ølnes et al., 2017). Case studies include real-estate transactions in Sweden, transferring property titles in the Republic of Georgia, and smart incorporation services in the US state of Delaware involving government agencies, companies, citizens and regulation bodies (Cheng et al., 2017). In the healthcare sector, blockchain has been shown to handle healthcare data management, including access control, data integrity, data sharing, and auditing, focusing on evaluating their trade-offs in terms of security, privacy, performance, and usability (Hardin and Kotz, 2019). For Kanagaretnam et al. (2018) trust underpins many financial processes.

Similarly, educational institutes have found that leveraging blockchain can help to manage the demand for human resources by streamlining the recruitment process; exchange records between institutions and monitor and evaluate the activities of students and staff by maintaining educational records (Yumna et al., 2019). Yessenbayev et al. (2024) consider applying blockchain and IoT to monitor radioactive waste information along its complete lifecycle for the purpose of managing nuclear waste. Li et al. (2023) examine the role of artificial intelligence in finance, focusing on its applications in financial forecasting and decision making. They suggest that integrating blockchain with artificial intelligence can enhance financial applications (see Li et al., 2023, p. 685). Within agriculture, blockchain has aided the food supply chain, improving food safety and transaction times (Bermeo-Almeida et al., 2018). While, in tourism, blockchain has operated hotel (Dong et al., 2020) and vehicle booking systems (Pallevada et al., 2021) for hospitality management, and transformed smart tourism (Bodkhe et al., 2019). Blockchain has been used together with other technologies to achieve smart public-safety systems that facilitate the exchange of open data while controlling data access permissions (Wang and Liu, 2023). Most notably, these applications have utilised several benefits of blockchain technology in public data: 1) providing trusted data storage and sharing, 2) simplifying data communication, 3) facilitating data access control, 4) enabling data auditing, verification, and traceability, and 5) eliminating intermediaries. These features are also applicable for beneficial ownership registration.

4.3. Digital identifiers

Another benefit of blockchain technology is the integration with corporate digital identity systems, which has implications for beneficial ownership transparency and financial market integrity. A corporate digital identifier strengthens fraud detection and reduces criminal and terrorist exploitation of the financial system (Leung et al., 2022). This goes beyond merely onboarding a corporate customer (FATF, 2020). Financial institutions utilise digital solutions to monitor real-time changes in the structures of legal entities, including ultimate beneficial ownership and directorships. This innovation transforms the AML/CTF landscape by reducing the exploitation of information asymmetries across jurisdictions.

Corporate digital identifiers operate in two modes: centralised and

decentralised. The decentralised model, often coordinated through DLTs, like blockchain, distributes identity data across multiple parties, offering flexibility and mitigating the risks associated with a single point of failure. A hybrid approach, combining elements of both centralised and decentralised systems, is emerging as a viable option that involves a single source of corporate identity, like a company registry, with various attributes and data linked in a network structure. This model mirrors individual identifier systems, like Singapore's MyInfo, and India's Aadhaar, where a single identity links to diverse golden source data, not aggregated at one point but linked as needed (Leung et al., 2022). Such a hybrid system, incorporating DLT, might enhance corporate digital identification by providing a more secure, efficient, and transparent method for managing and verifying corporate and beneficial ownership information.

Insights can also be drawn from the implementation of Estonia's e-card digital identity system. An interoperability platform, X-Road, developed in Estonia in 2001, is now in use across Finland, Azerbaijan, Namibia, Faroe Islands and Mexico (PwC, 2019; Sullivan and Tyson, 2023). X-Road facilitates the automatic exchange of data across jurisdictions and enables real-time verification of government records, improving the management of public sector systems, such as Estonia's property and business registries (PwC, 2019, p. 7). Estonia's e-card system utilises blockchain's public key infrastructure and government-issued digital certificates containing residents' biometric data, including name, date of birth, and personal identifiers, to authenticate both users and their devices (Idefy, 2023). The system has streamlined commercial transactions, allowing Estonians to register company data, open bank accounts, transfer money, and trade goods and services (Sullivan and Burger, 2017). Its success lies in its ability to reduce fraud, while enabling citizens to remotely access government and private sector services (Idefy, 2023). However, as Sullivan and Tyson (2023, pp. 441–442) demonstrate, the use of blockchain technology for digital identifiers remains largely untested. A key challenge is determining who should be responsible for verifying information contained within such a system. To address this, Sullivan and Tyson (2023) propose a hybrid approach that integrates distributed ledgers with existing Know-Your-Customer systems and established legal and procedural standards. This approach may also assist accountants and other regulated professionals in verifying beneficial ownership.

Digitising corporate ownership data aids AML/CTF efforts by providing timely access to ownership details—decisive for AML/CTF teams in financial institutions (Monroe, 2023). Some countries, like Denmark, Hong Kong, Belgium, Austria, and the Netherlands, have implemented cross-check systems that automatically verify information against other national databases, such as address and identification registers, which helps to confirm the identities of listed individuals (Turner and Evstropova, 2020). Such a system might ensure each entry about a company's beneficial owner held in the register has a unique digital identifier that would help to trace beneficial ownership and identify ownership patterns or links between companies (Gilmour, 2022a; OpenOwnership and Global Witness, 2017; Lemieux, 2016).

Nonetheless, this necessitates developing interoperability between different blockchains to facilitate functionalities, such as cross-chain payment. These cross-chain approaches, not only, facilitate the transfer of information between blockchains, but also, enhance blockchain performance by distributing transaction verification tasks across multiple chains. Such an ecosystem could comprise various chains, each serving distinct purposes and involving only its relevant stakeholders. An ecosystem could be built with various chains each serving different purposes only evolving its relevant stakeholders. Notably, the static nature of validator sets in some permissioned blockchains raises concerns about collusion and improper selection, which compromise system trust (Viriyasitavat et al., 2019, p. 743). Viriyasitavat et al. (2019) suggest integrating service-oriented architecture and public-key infrastructure with blockchain to enhance flexibility, validate data, and certify validator sets, aiming to ensure reliable interoperations within

Internet of Things (IoT) environments. Blockchain may provide for an interoperable and consistent register of data by integrating blockchain without disturbing existing data systems and business processes (Shahaab et al., 2023).

5. Balancing the need for privacy and enhanced transparency

While the role of technology in maintaining registry information is progressing, questions are being raised about the reliability of registry data, access to sensitive data on beneficial ownership and related data privacy (Gilmour, 2020; Ratcliffe, 2023; Thomas-James, 2023). Liyanaarachchi et al. (2024) note the emerging 'privacy paradox', wherein blockchain's inherent transparency clashes with the need to comply with data protection laws. This paradox is amplified by growing societal resistance to constraints on consumer freedom and concerns over unauthorised data access. Notably, public access to beneficial ownership registries continues to be subject of great debate: balancing the deterrence against criminal misuse with the privacy rights of beneficial owners (Ratcliffe, 2023). These debates around the reliability of data registry and reporting discrepancies (see Matras, 2023) along with balancing the public access and privacy rights, can be mitigated through blockchain. According to Liyanaarachchi et al. (2024) a modified blockchain with selective immutability, federal decentralisation, and supervised transparency may overcome the 'privacy paradox' and offer a solution to data-privacy concerns. Leveraging blockchain, records are preserved on a public blockchain, ensuring its immutability and verifiability, diminishing the likelihood of tampering due to corruption and political interferences (Natarajan et al., 2017). The complexity between transparency and security in beneficial ownership registries can correspond with blockchain's inherent decentralisation, immutability, and tamper-proof mechanism suitable for the management of beneficial ownership data.

Privacy is a concern, especially in jurisdictions with personal safety risks, such as Mexico (Kiepe, 2021). The balance between transparency and privacy protection is questioned in implementing these registers. The lack of data verification resources and varying approaches to registry accessibility contribute to difficulties in ensuring data accuracy. In India, for instance, there is a complexity in identifying significant beneficial owners in layered corporate structures and a need for more transparent mechanisms (Pai, 2023). Furthermore, the issues in updating and capturing registry data are also prominent (Monroe, 2023).

Governments are also carefully considering the implications of the ruling by the Court of Justice of the European Union (CJEU). In November 2022, the CJEU considered a joint case brought against the Luxembourg Business Registers by two companies seeking to prevent information about their beneficial owners from becoming publicly available (See *Court of Justice of the European Union Grand Chamber*, 2022). The CJEU ruled that it was disproportionate and unlawful for EU Member States to make beneficial ownership data accessible in all cases to the 'general public' (Norton Rose Fulbright, 2023; Thomas-James, 2023). The CJEU considered that sensitive personal information held within register systems must be protected from potential abuse and accessed in an accountable and proportionate manner. The judgement effectively invalidates aspects of the EU's 5th Anti-Money Laundering Directive that guaranteed full transparency of beneficial ownership. Consequently, some jurisdictions (Austria, Belgium, Cyprus, and Germany to name a few) have opted to limit public access or allow beneficial owners to request the removal of their information because of clarity following the CJEU ruling (Lorenzo, 2023; Matras, 2023). Therefore, accountants must also carefully align their AML efforts with protecting citizens' fundamental rights to privacy.

Finally, we recognise that the decentralised nature of blockchain removes central control, which can result in regulatory fragmentation (Li, 2021; Weber, 2025). This might serve to undermine public trust and safety until such technology can be fully integrated into existing regulatory frameworks with proper oversight, accountability and

governance structures. Furthermore, blockchains rely on cryptographic consensus algorithms that have embedded biases and can perpetuate discrimination based on race, gender, or socio-economic status (see Hussein et al., 2023). Algorithmic discrimination could lead to unfair practices, where individuals or groups are marginalised or excluded from economic opportunities due to biases built into the code (Ghasemaghaei and Nima Kordzadeh, 2024; Li, 2021). Such biases must be addressed through a risk-avoidance framework for developers, companies, and governments that ensures moral responsibility. This means implementing ethical standards whereby algorithms are transparent and auditable, and regular audits of decision-making processes.

6. Conclusion

As research into blockchain evolves, a clearer understanding about the issues of adopting new technologies for accounting purposes emerges. Thus far, blockchain research has mainly been conceptual based (Shahaab et al., 2023) or centred on proposed solutions to existing real-world issues (Gilmour, 2022a; Pimentel and Boulianne, 2020). We agree that research should focus on examining the issues that the accounting sector faces in adopting blockchain technology. Such an approach will help investigate what works on a practical level and to highlight best practices. It is important to focus blockchain research on accounting practice to bridge the gap between theoretical concepts and real-world implementation to pave the way for a more nuanced understanding of blockchain's adoption in registers of beneficial owners.

This paper has examined the role of blockchain technology within registers of beneficial owners and highlighted key implications for the accounting sector. We have studied how the disclosure of beneficial ownership operates to enhance the transparency of companies through analysing some existing centralised registers. We acknowledge that the idea of centralised registers of beneficial owners is one that has merit for combating corporate secrecy, but how centralised registers are currently implemented is proving problematic.

Blockchain technology is a novel digital technology that serves as an ideal trust machine (Miraz and Ali, 2018) yet it must ensure transparency and access for accountants in providing the digital infrastructure needed to benefit registers of beneficial owners. This paper positions accountants as central gatekeepers in AML/CTF compliance and in the global fight against financial crime. In doing so, we introduce a novel conceptual model to represent how trust, transparency and accessibility are crucial determinants for an effective blockchain-based system for enhancing beneficial ownership that accountants can rely on.

It is likely that researchers will need to overcome several key challenges in order to successfully adopt blockchain and meet the needs of the accounting sector. This is in addition to the issues of verification, trust and data privacy as discussed already. First, enhancing interoperability between blockchain platforms may prove difficult as each blockchain operates within its own defined set of protocols, restricting how they interact with one another (El Koshiry et al., 2023). Further work is needed to ensure the growing data on beneficial owners and their companies are exchanged and shared across platforms in a seamless manner to encourage responsible risk-based decisions and engagement with the system (Shahaab et al., 2023). Second and relatedly, the issues of scalability of such data systems must be explored to enable mass data on beneficial ownership to be managed by relevant stakeholders, including law enforcement, banks, accountants, and other appropriate stakeholders (El Koshiry et al., 2023). This might include implementing a modified blockchain and exploring how new processes that involve blockchain technology become integrated with existing processes or systems, while minimising costs and maintaining the security of beneficial owners' personal data (El Koshiry et al., 2023; Liyanaarachchi et al., 2024; Sousa, 2023; Sullivan and Burger, 2017; Sullivan and Tyson, 2023). As Li et al. (2023, p. 685) suggest, artificial intelligence and machine-learning tools can be integrated with blockchain to help solve such problems (see also, Chen et al., 2023).

Nonetheless, given blockchain's relative infancy as a digital technology (Attaran, 2022; El Koshiry et al., 2023) further research is needed to overcome the barriers to adopting blockchain for the accounting profession.

CRedit authorship contribution statement

Paul Gilmour: Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization. **Durgesh Pandey:** Writing – review & editing, Writing – original draft. **Doron Goldbarsht:** Writing – review & editing, Writing – original draft.

Ethics

We confirm that this work is original and has neither been published nor is currently considered for publication elsewhere. We confirm that each of the co-authors acknowledges their participation in conducting the research leading to this manuscript and has agreed to its submission to be considered for publication. The authors declare that no experimental procedures were conducted in which animals/human participation/trial were involved. We have prepared the manuscript including figure and tables in accordance with the journal's style and format requirements.

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Data availability

No data was used for the research described in the article.

References

- Acharya, V.V., Lambrecht, B.M., 2015. A theory of income smoothing when insiders know more than outsiders. *Rev. Financ. Stud.* 28 (9), 2534–2574. <https://doi.org/10.1093/rfs/hhv026>.
- Advani, A., Poux, C., Powell-Smith, A., Summers, A., 2023. Catch me if you can: gaps in the register of overseas entities. available at: https://warwick.ac.uk/fac/soc/economics/research/centres/cage/publications/workingpapers/2023/catch_me_if_you_can_gaps_in_the_register_of_overseas_entities.
- Albayati, H., Kim, S.K., Rho, J.J., 2020. Accepting financial transactions using blockchain technology and cryptocurrency: A customer perspective approach. *Technol. Soc.* 62 (1). <https://doi.org/10.1016/j.techsoc.2020.101320>.
- Angluin, D., Scapens, R.W., 2000. Transparency, accounting knowledge and perceived fairness in UK universities' resource allocation: results from a survey of accounting and finance. *Br. Account. Rev.* 32 (1), 1–42. <https://doi.org/10.1006/bare.1999.0119>.
- Attaran, M., 2022. Blockchain technology in healthcare: challenges and opportunities. *Int. J. Healthc. Manag.* 15 (1), 70–83. <https://doi.org/10.1080/20479700.2020.1843887>.
- Azrilyant, J., 2021. Shell game: how the corporate transparency act aims to end the illicit use of shell companies, where it fails, and what to do about it. *Public Contract Law J.* 51 (1), 1–27.
- Baudet, C., Medina, M.J., 2023. The paradoxes of trust and transparency of blockchain technologies. *J. Glob. Inf. Manag.* 31 (5), 1–22. <https://doi.org/10.4018/JGIM.321111>.
- Bermeo-Almeida, O., Cardenas-Rodriguez, M., Samaniego-Cobo, T., Ferruzola-Gómez, E., Cabezas-Cabezas, R., Bazán-Vera, W., 2018. Blockchain in agriculture: a systematic literature review. In: *Technologies and Innovation: 4th International Conference, CITI 2018, Proceedings 4*. Springer, Cham, pp. 44–56. https://doi.org/10.1007/978-3-030-00940-3_4.
- Bodkhe, U., Bhattacharya, P., Tanwar, S., Tyagi, S., Kumar, N., Obaidat, M.S., 2019. August. BloHosT: Blockchain enabled smart tourism and hospitality management. In: *2019 International Conference on Computer, Information and Telecommunication Systems (CITS) International Conference on Computer, Information and Telecommunication Systems (CITS)*. IEEE, pp. 1–5. <https://doi.org/10.1109/CITS.2019.8862001>.

- Capocasale, V., Gotta, D., Perboli, G., 2022. Comparative analysis of permissioned blockchain frameworks for industrial applications. *Blockchain: Res. Appl.* 4 (1), 100113. <https://doi.org/10.1016/j.bcr.2022.100113>.
- Casino, F., Dasaklis, T.K., Patsakis, C., 2019. A systematic literature review of blockchain-based applications: current status, classification and open issues. *Telemat. Inform.* 36, 55–81. <https://doi.org/10.1016/j.tele.2018.11.006>.
- Centobelli, P., Cerchione, R., Del Vecchio, P., Oropallo, E., Secundo, G., 2022. Blockchain technology for bridging trust, traceability and transparency in circular supply chain. *Inf. Manag.* 59 (7), 1–14. <https://doi.org/10.1016/j.im.2021.103508>.
- Chen, H., 2023. Blockchain technology and small and medium enterprises access to finance. *Adv. Econ. Manag. Political Sci.* 8, 138–144. <https://doi.org/10.54254/2754-1169/8/20230297>.
- Chen, W., He, W., Shen, J., Tian, X., Wang, X., 2023. Systematic analysis of artificial intelligence in the era of industry 4.0. *J. Manag. Anal.* 10 (1), 89–108. <https://doi.org/10.1080/23270012.2023.2180676>.
- Cheng, S., Daub, M., Domesy, A., Lundqvist, M., 2017. Using blockchain to improve data management in the public sector. available at: <http://dln.jaipuria.ac.in:8080/js-pui/bitstream/123456789/2925/1/Using-blockchain-to-improve-data-management-in-the-public-sector.pdf>. accessed 20.12.2023.
- Clarkson, G., Jacobsen, T.E., Batcheller, Archer L., A. L., 2007. Information asymmetry and information sharing. *Gov. Inf. Q.* 24 (4), 827–839. <https://doi.org/10.1016/j.giq.2007.08.001>.
- Cobham, A., 2013. July 23. Corrupting perceptions: why transparency international's flagship corruption index falls short. available at: <https://www.cgdev.org/blog/corrupting-perceptions-why-transparency-international%E2%80%99s-flagship-corruption-index-falls-short>. accessed 30.12.2023.
- Companies House, 2023. Companies House: Guidance for limited companies, partnerships and other company types. available at: <https://www.gov.uk/government/collections/companies-house-guidance-for-limited-companies-partnerships-and-other-company-types>. accessed 02.01.2024.
- Court of Justice of the European Union Grand Chamber, 22 November 2022. Joined Cases C-37/20 and C-601/20, WM and Sovim. SA v Luxembourg Business Registers. available at: accessed 18.01.2024 <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:62020CJ0037>.
- Dai, J., Vasarhelyi, M.A., 2017. Toward blockchain-based accounting and assurance. *J. Inf. Syst.* 31 (3), 5–21. <https://doi.org/10.2308/isys-51804>.
- Delmotte, C.J., 2023. Toward a blockchain-driven tax system. *Va. Tax Rev.* 43 (1), 37–86.
- Dentons, 2018, September 18. Practical application of distributed ledger technology: Maintaining corporate records on the blockchain. available at: <https://www.denton.com/en/insights/articles/2018/september/28/practical-application-of-distributed-ledger-technology>. accessed 20.11.2023.
- Dong, N., Bai, G., Huang, L.-C., Lim, E., Kok, H., Dong, J.S., 2020. A blockchain-based decentralized booking system. *Knowl. Eng. Rev.* 35, e17. <https://doi.org/10.1017/s026988920000260>.
- El Koshiry, A., Eliwa, E., Abd El-Hafeez, T., Shams, M.Y., 2023. Unlocking the power of blockchain in education: an overview of innovations and outcomes. *Blockchain: Res. Appl.* 4 (4). <https://doi.org/10.1016/j.bcr.2023.100165>.
- Ertan, A., Loumisti, M., Wittenberg-Moerman, R., 2017. Enhancing loan quality through transparency: evidence from the European Central Bank loan level reporting initiative. *J. Account. Res.* 55 (4), 877–918. <http://www.jstor.org/stable/26650977>.
- Financial Action Task Force, 1992. FATF annual report 1991–1992. available at: <https://www.fatf-gafi.org/en/publications/Fatfgeneral/Fatfannualreport1991-1992.html>. accessed 18.01.2024.
- Financial Action Task Force, 2002. FATF annual report 2001–2002. available at: <https://www.fatf-gafi.org/en/publications/Fatfgeneral/Fatfannualreport2001-2002.html>. accessed 18.01.2024.
- Financial Action Task Force, 2003. FATF annual report 2002–2003. available at: <https://www.fatf-gafi.org/en/publications/Fatfgeneral/Fatfannualreport2002-2003.html>.
- Financial Action Task Force, 2020. Guidance on digital identity. available at: accessed 18.01.2024 www.fatf-gafi.org/publications/documents/digital-identity-guidance.html.
- Financial Action Task Force, 2022. Report on the state of effectiveness and compliance with the FATF standards. available at: <https://www.fatf-gafi.org/en/publications/Fatfgeneral/Effectiveness-compliance-standards.html>. accessed 20.11.2023.
- Financial Action Task Force, 2023a. Guidance on beneficial ownership for legal persons. available at: <http://www.fatf-gafi.org/publications/FATFrecommendations/guidance-beneficial-ownership-legalpersons.html>. accessed 15.11.2023.
- Financial Action Task Force, 2023b. International standards on combating money laundering and the financing of terrorism & proliferation the FATF recommendations. available at: <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Fatf-recommendations.html>. accessed 06.01.2024.
- Gan, Q., Lau, R.Y.K., 2024. Trust in a 'trust-free' system: Blockchain acceptance in the banking and finance sector. *Technol. Forecast. Soc. Change* 199, 123050. <https://doi.org/10.1016/j.techfore.2023.123050>.
- Gardner, T.A., Benzie, M., Börner, J., Dawkins, E., Fick, S., Garrett, R., Godar, J., Grimard, A., Lake, S., Larsen, R.K., Mardas, N., McDermott, C.L., Meyfroidt, P., Osbeck, M., Persson, M., Sembres, T., Suavet, C., Strassburg, B., Trevisan, A., West, C., Wolvekamp, P., 2019. Transparency and sustainability in global commodity supply chains. *World Dev.* 121, 163–177. <https://doi.org/10.1016/j.worlddev.2018.05.025>.
- Garg, D., Alam, M., 2023. Smart agriculture: a literature review. *J. Manag. Anal.* 10 (2), 359–415. <https://doi.org/10.1080/23270012.2023.2207184>.
- Gaur, V., 2023, August 29. Understanding beneficial ownership in the Indian corporate landscape. available at: <https://www.cbfinludelihi.in/post/understanding-beneficial-ownership-in-the-indian-corporate-landscape>. accessed 07.01.2024.
- Ghasemaghahi, M., Nima Kordzadeh, N., 2024. Understanding how algorithmic injustice leads to making discriminatory decisions: an obedience to authority perspective. *Inf. Manag.* 61 (2), 103921. <https://doi.org/10.1016/j.im.2024.103921>.
- Gillies, A., Heuty, A., 2011. Does transparency work? The challenges of measurement and effectiveness in resource-rich countries. *Yale J. Int. Aff.* 6 (2), 25–42.
- Gillis, M., 2019. Shell companies and exposing beneficial ownership: testing the boundaries of the international commitment to fight corruption. *Melb. J. Int. Law* 20 (2), 378–400. https://law.unimelb.edu.au/_data/assets/pdf_file/0011/3567440/Gillis.pdf.
- Gilmour, P.M., 2020. Lifting the veil on beneficial ownership: challenges of implementing the UK's registers of beneficial owners. *J. Money Laund. Control* 23 (4), 717–734. <https://doi.org/10.1108/JMLC-02-2020-0014>.
- Gilmour, P.M., 2022a. Decentralized blockchain technology: Towards a trusted and transparent beneficial ownership regime. In: Goldbarsht, D., de Koker, L. (Eds.), *Financial Technology and the Law: Combating Financial Crime, Law, Governance and Technology Series*, vol. 47. Springer, pp. 185–209. <https://doi.org/10.1007/978-3-030-88036-1>.
- Gilmour, P.M., 2022b. The US Corporate Transparency Act: critiquing beneficial ownership disclosure requirements in the United States. *Company Lawyer* 43 (1).
- Gilmour, P., 2023. Smart contracts and the metaverse. *Company Lawyer* 44 (7), 244–245.
- Global Witness, 2018. The Companies we keep: what the UK's open data register actually tells us about company ownership. available at: <https://www.globalwitness.org/en/campaigns/corruption-and-money-laundering/anonymous-company-owners/companies-we-keep/>. accessed 15.11.2023.
- Goergen, M., Mira, S., 2023. Measuring the ownership and control of UK listed firms: some methodological challenges. *Br. Account. Rev.* 55 (6), 101272. <https://doi.org/10.1016/j.bar.2023.101272>.
- Goldbarsht, D., 2017. Who's the legislator anyway? How the FATF's global norms reshape Australian counter terrorist financing laws. *Fed. Law Rev.* 45 (1), 127–151. <https://doi.org/10.1177/0067205X1704500>.
- Goldbarsht, D., 2020. Reverse engineering legal professional privilege in a globalising world – the Australian case. *J. Money Laund. Control* 23 (3), 677–690. <https://doi.org/10.1108/JMLC-02-2020-0011>.
- Goldbarsht, D., Benson, K., 2023. From later to sooner: exploring compliance with the global regime of anti-money laundering and counter-terrorist financing in the legal profession. *J. Financ. Crime* 31 (4), 795–809. <https://doi.org/10.1108/JFC-08-2023-0201>.
- Guggenberger, T., Sedlmeir, J., Fridgen, G., Luckow, A., 2022. An in-depth investigation of the performance characteristics of Hyperledger Fabric. *Comput. Ind. Eng.* 173, 108716. <https://doi.org/10.1016/j.cie.2022.108716>.
- Guo, Y., Liang, C., 2016. Blockchain application and outlook in the banking industry. *Financ. Innov.* 2, 24. <https://doi.org/10.1186/s40854-016-0034-9>.
- Hamidi, S.M.M., Hoseini, S.F., Gholami, H., Kananzadeh-Bahmani, M., 2024. A three-stage digital maturity model to assess readiness for blockchain implementation in the maritime logistics industry. *J. Ind. Inf. Integr.* 41, 100643. <https://doi.org/10.1016/j.jii.2024.100643>.
- Hardin, T., Kotz, D., 2019. Blockchain in health data systems: a survey. In: *In 2019 Sixth International Conference on Internet of Things: Systems, Management and Security (IOTSMS)*. IEEE, pp. 490–497.
- Hussein, Z., Salama, M.A., El-Rahman, S.A., 2023. Evolution of blockchain consensus algorithms: A review on the latest milestones of blockchain consensus algorithms. *Cybersecurity* 6 (30). <https://doi.org/10.1186/s42400-023-00163-y>.
- Idefy, 2023, October 31. Estonia's blockchain-based digital identity system: A model for the world. available at: <https://idefy.ai/estonia-blockchain-based-digital-identity-system-a-model-for-the-world>. accessed 09.02.2025.
- Institute of International Finance, 2017. Deploying Regtech against financial crime. available at: https://www.iif.com/portals/0/Files/Private/32370132_aml_final_id.pdf. accessed 19.11.2023.
- Javaid, M., Haleem, A., Singh, R.P., Suman, R., Khan, S., 2022. A review of blockchain technology applications for financial services. *BenchCouncil Trans. Benchmarks Stand. Eval.* 2 (3), 100073. <https://doi.org/10.1016/j.tbench.2022.100073>.
- Jeanneret Medina, M., Baudet, C., Lebraty, J.-F., 2024. Blockchain and agency theory in supply chain management: A question of trust. *Int. J. Inf. Manag.* 75, 102747. <https://doi.org/10.1016/j.ijinfomgt.2023.102747>.
- Kabir, M.R., 2021. Behavioural intention to adopt blockchain for a transparent and effective taxing system. *J. Glob. Oper. Strateg. Sourc.* 14 (1), 170–201. <https://doi.org/10.1108/JGOSS-08-2020-0050>.
- Kanagaretnam, K., Lee, J., Lim, C.Y., Lobo, G., 2018. Societal trust and corporate tax avoidance. *Rev. Acc. Stud.* 23, 1588–1628. <https://doi.org/10.1007/s11142-018-9466-y>.
- Khan, S.N., Loulik, F., Ghedira-Guegan, C., Benkhelifa, E., Bani-Hani, A., 2021. Blockchain smart contracts: applications, challenges, and future trends. *Peer Peer Netw Appl* 14, 2901–2925. <https://doi.org/10.1007/s12083-021-01127-0>.
- Khawaja, S., Javidroozi, V., 2023. Blockchain technology as an enabler for cross-sectoral systems integration for developing smart sustainable cities. *IET Smart Cities* 5 (3), 151–172. <https://doi.org/10.1049/smc2.12059>.
- Kiepe, T., 2021, May 25. Making central beneficial ownership registers public. Available at Open Ownership: <https://www.openownership.org/en/publications/making-central-beneficial-ownership-registers-public/overview>. accessed 19.12.2023.
- Kim, K., Kauffman, R.J., 2024. On the effects of information asymmetry in digital currency trading. *Electron. Commer. Res. Appl.* 64, 101366. <https://doi.org/10.1016/j.elerap.2024.101366>.

- Knobel, A., 2019, January 20. Beneficial ownership verification: ensuring the truthfulness and accuracy of registered ownership information. available at tax justice network: <https://taxjustice.net/reports/beneficial-ownership-verification-ensuring-the-truthfulness-and-accuracy-of-registered-ownership-information>. accessed 30.12.2023.
- Knobel, A., 2023, December 20. New report on how to fix beneficial ownership frameworks, so they actually work. available at Tax Justice Network: <https://taxjustice.net/2023/12/20/new-report-on-how-to-fix-beneficial-ownership-frameworks-so-they-actually-work>. accessed 30.12.2023.
- Knobel, A., Lorenzo, F., 2022. Beneficial ownership registration around the world 2022. available at Tax Justice Network: <https://taxjustice.net/wp-content/uploads/2022/12/State-of-Play-of-Beneficial-Ownership-2022-Tax-Justice-Network.pdf>. accessed 19.12.2023.
- Kononova, M., Tuck, P., Ormeño Pérez, R., 2023. In search of the owner: regulating through transparency. *Crit. Perspect. Account.* 91, 102421. <https://doi.org/10.1016/j.cpa.2022.102421>.
- Kuo, T.-T., Kim, H.-E., Ohno-Machado, L., 2017. Blockchain distributed ledger technologies for biomedical and health care applications. *J. Am. Med. Inform. Assoc.* 24 (6), 1211–1220. <https://doi.org/10.1093/jamia/ocx068>.
- Kyles, D.L., 2022. Centralised control over decentralised structures: AML and CTF regulation of blockchains and distributed ledgers. In: Goldbarsht, D., de Koker, L. (Eds.), *Financial Technology and the Law: Combating Financial Crime, Law, Governance and Technology Series*, vol. 47. Springer, pp. 121–150. <https://doi.org/10.1007/978-3-030-88036-1>.
- Lemieux, V.L., 2016. Trusting records: is blockchain technology the answer? *Rec. Manag. J.* 26 (2), 110–139. <https://doi.org/10.1108/RMJ-12-2015-0042>.
- Leung, D., Nolens, B., Arner, D., Frost, J., 2022, June 16. Corporate digital identity: no silver bullet, but a silver lining. Basel: Bank for International Settlements. available at: <https://www.bis.org/publ/bppdf/bispap126.htm>. accessed 19.12.2023.
- Li, X., Sigov, A., Ratkin, M., Ivanov, L.A., Li, L., 2023. Artificial intelligence applications in finance: A survey. *J. Manag. Anal.* 10 (4), 676–692. <https://doi.org/10.1080/23270012.2023.2244503>.
- Li, Z., 2021. Confronting algorithm bias risks: will blockchain provide new opportunities or challenges for data protection law? *Dublin Law Politics Rev.* 2, 43–67 (Finance Special Issue).
- Littrell, C., 2024. Biases in FATF mutual evaluation results. In: Goldbarsht, D., de Koker, L. (Eds.), *Financial Crime and the Law: Identifying and Mitigating Risks*. Springer.
- Liyaanarachchi, G., Viglia, G., Kurtaliqui, F., 2024. Addressing challenges of digital transformation with modified blockchain. *Technol. Forecast. Soc. Change* 201, 123254. <https://doi.org/10.1016/j.techfore.2024.123254>.
- de Long, J., Meyer, A., Owens, J., 2017. Using blockchain for transparent beneficial ownership registers. *Int. Tax Rev.* 39, 2–5.
- Lorenzo, F., 2023, July 13. Split among EU countries over beneficial ownership ruling mirrors rankings on Financial Secrecy Index. available at: Tax Justice Network: <https://taxjustice.net/2023/07/13/split-among-eu-countries-over-beneficial-ownership-ruling-mirrors-rankings-on-financial-secrecy-index>. accessed 03.01.2024.
- Lu, Y., 2019. Artificial intelligence: A survey on evolution, models, applications and future trends. *J. Manag. Anal.* 6 (1), 1–29. <https://doi.org/10.1080/23270012.2019.1570365>.
- Luo, M., Yu, S., 2022. Financial reporting for cryptocurrency. *Rev. Acc. Stud.* <https://doi.org/10.1007/s11442-022-09741-w>.
- Manganaris, P., Beccalli, E., Dimitropoulos, P., 2017. Bank transparency and the crisis. *Br. Account. Rev.* 49 (2), 121–137. <https://doi.org/10.1016/j.bar.2016.07.002>.
- Marella, V., Upreti, B., Merikivi, J., Tuunainen, V.K., 2020. Understanding the creation of trust in cryptocurrencies: the case of bitcoin. *Electron. Mark.* 30 (2), 259–271. <https://doi.org/10.1007/s12525-019-00392-5>.
- Markle, A., Kiepe, T., 2022. Who benefits? How company ownership data is used to detect and prevent corruption. Open Ownership: accessed 06.01.2023. <https://www.openownership.org/en/publications/who-benefits-how-company-ownership-data-is-used-to-detect-and-prevent-corruption>.
- Matras, T.M., 2023. Functioning of the register of beneficial owners: findings from Poland. *J. Money Laund. Control* 27 (5), 932–943. <https://doi.org/10.1108/JMLC-09-2023-0149>.
- Miraz, M., Ali, M., 2018. Applications of blockchain technology beyond cryptocurrency. *Ann. Emerg. Technol. Comput. (AETIC)* 2 (1), 1–6. <https://doi.org/10.33166/AETIC.2018.01.001>.
- Monroe, B., 2023, May 18. In quest to capture data, conquer corporate ownership registry challenges, Moody's Analytics' Ljubinka Slavenska gets 'absorbed' into AML world. available at ACFCs: <https://www.acfcs.org/in-quest-to-capture-data>. accessed 19.12.2023.
- Morgan, E., 2022. All you need to know about the UK Trust Registration Service. available at The Gazette: <https://www.thegazette.co.uk/wills-and-probate/content/104031>. accessed 02.01.2024.
- Nakamoto, S., 2008. Bitcoin: a peer-to-peer electronic cash system. available at United States Sentencing Commission: https://www.uscc.gov/sites/default/files/pdf/training/annual-national-training-seminar/2018/Emerging_Tech_Bitcoin_Crypto.pdf. accessed 31.12.2023.
- Natarajan, H., Solvej, Krause S., Gradstein, H., 2017. Distributed ledger technology (DLT) and blockchain. International Bank for Reconstruction and Development and World Bank, available at: <https://documents1.worldbank.org/curated/en/177911513714062215/pdf/122140-WP-PUBLIC-Distributed-Ledger-Technology-and-Blockchain-Fin-tech-Notes.pdf>. accessed 21.11.2023.
- Nauli, P., Mohamed, N., Sultan, N., 2023. From secrecy to accountability: the evolution of beneficial ownership practices in Indonesia's anti-money laundering framework. *Pak. J. Criminol.* 15 (4), 611–637.
- Navadkar, V.H., Nighot, A., Wantmure, R., 2018. Overview of blockchain technology in government/public sectors. *Int. Res. J. Eng. Technol.* 5 (6), 2287–2292.
- Niu, H., Li, T., Gong, X., 2022. A blockchain-based certifiable anonymous E-taxing protocol. *PLOS One* 17 (7), e0270454. <https://doi.org/10.1371/journal.pone.0270454>.
- Norton Rose Fulbright, 2023. Regulation around the world: Beneficial ownership registers. available at: https://www.nortonrosefulbright.com/-/media/files/nrf/nrfweb/publications/v3-50108_emea_brochure_regulation-around-the-world-beneficial-ownership-registries.pdf. accessed 18.11.2023.
- Ølne, S., Ubacht, J., Janssen, M., 2017. Blockchain in government: benefits and implications of distributed ledger technology for information sharing. *Gov. Inf. Q.* 34 (3), 355–364. <https://doi.org/10.1016/j.giq.2017.09.007>.
- OpenOwnership and Global Witness, 2017. Learning the lessons from the UK's public beneficial ownership register. available at: https://www.globalwitness.org/documents/19250/Learning_the_Lessons_from_the_UKs_public_Beneficial_Ownership_register.pdf.
- Pai, V., 2023, March 31. PE funds face challenges identifying beneficial owners, India Business Law Journal. available at: <https://law.asia/pe-funds-challenges-identifying-g-sbe>. accessed 19.12.2023.
- Pallevada, H., Kanuri, G.P.K., Posina, S., Paruchuri, S., Chinta, M., 2021, October. Blockchain based decentralized vehicle booking service. In: In 2021 2nd International Conference on Smart Electronics and Communication (ICOSEC). IEEE, pp. 1418–1424.
- Park, Y.J., Chung, J.E., Shin, D.H., 2018. The structuration of digital ecosystem, privacy, and big data intelligence. *Am. Behav. Sci.* 62 (10), 1319–1337. <https://doi.org/10.1177/0002764218787863>.
- Pimentel, E., Boulianne, E., 2020. Blockchain in accounting research and practice: current trends and future opportunities. *Account. Perspect.* 19 (4), 325–361. <https://doi.org/10.1111/1911-3838.12239>.
- PwC, 2019. Estonia—the digital republic secured by blockchain. available at: <https://www.pwc.com/gx/en/services/legal/tech/assets/estonia-the-digital-republic-secured-by-blockchain.pdf>. accessed 09.02.2025.
- Ranaldo, A., Somogyi, F., 2021. Asymmetric information risk in FX markets. *J. Financ. Econ.* 140 (2), 391–411. <https://doi.org/10.1016/j.jfineco.2020.12.007>.
- Rane, S.B., Thakker, S.V., Kant, R., 2021. Stakeholders' involvement in green supply chain: a perspective of blockchain IoT-integrated architecture. *Manag. Environ. Qual. Int. J.* 32 (6), 1166–1191. <https://doi.org/10.1108/MEQ-11-2019-0248>.
- Ratcliffe, H., 2023, May 3. Divided opinion: The continuing debate on public access to registers of beneficial ownership. available at: <https://www.ifcreview.com/article/s/2023/may/divided-opinion-the-continuing-debate-on-public-access-to-registers-of-beneficial-ownership>. accessed 03.01.2024.
- Rico-Peña, J.J., Arguedas-Sanz, R., López-Martin, C., 2023. Models used to characterise blockchain features. A systematic literature review and bibliometric analysis. *Technovation* 123, 102711. <https://doi.org/10.1016/j.technovation.2023.102711>.
- Robbins, G., Lapsley, I., 2015. From secrecy to transparency: accounting and the transition from religious charity to publicly-owned hospital. *Br. Account. Rev.* 47 (1), 19–32. <https://doi.org/10.1016/j.bar.2014.10.008>.
- Sahut, J.-M., Schweizer, D., Peris-Ortiz, M., 2022. Technological forecasting and social change introduction to the VSI technological innovations to ensure confidence in the digital world. *Technol. Forecast. Soc. Change* 179, 121680. <https://doi.org/10.1016/j.techfore.2022.121680>.
- Segura Ibarra, D., Li, F., Jianjun Zhu, J., Chen, J., 2024. Blockchain application to the processes in material design, production, distribution, and disposal: A survey. *J. Ind. Inf. Integr.* 41, 100638. <https://doi.org/10.1016/j.jii.2024.100638>.
- Shahaab, A., Khan, I.A., Maude, R., Hewage, C., Wang, Y., 2023. Public service operational efficiency and blockchain: A case study of companies house, UK. *Gov. Inf. Q.* 40 (1), 101759. <https://doi.org/10.1016/j.giq.2022.101759>.
- Shin, D.D.H., 2019. Blockchain: the emerging technology of digital trust. *Telemat. Inform.* 45, 101278. <https://doi.org/10.1016/j.tele.2019.101278>.
- Sihombing, J.A., 2023. Beneficial ownership: transparency as an effort to prevent and eradicate money laundering and its impact on investment markets. *Policy, Law, Notary Regul. Issues* 2 (3), 208–218. <https://doi.org/10.55047/polri.v2i3.642>.
- Simser, J., 2015. Bitcoin and modern alchemy: in code we trust. *J. Financ. Crime* 22 (2), 156–169. <https://doi.org/10.1108/JFC-11-2013-0067>.
- Sousa, M.J., 2023. Blockchain as a driver for transformations in the public sector. *Policy Des. Pract.* 6 (4), 415–432. <https://doi.org/10.1080/25741292.2023.2267864>.
- Sullivan, C., Burger, E., 2017. E-residency and blockchain. *Comput. Law Secur. Rev.* 33 (4), 470–481. <https://doi.org/10.1016/j.clsr.2017.03.016>.
- Sullivan, C., Tyson, S., 2023. A global digital identity for all: the next evolution. *Policy Des. Pract.* 6 (4), 433–445. <https://doi.org/10.1080/25741292.2023.2267867>.
- Thomas-James, D., 2022. Editorial: interdicting economic crime through beneficial ownership: an international standard at last? *J. Money Laund. Control* 25 (3), 489–492. <https://doi.org/10.1108/JMLC-05-2022-150>.
- Thomas-James, D., 2023. Editorial: the court of justice of the European Union and the beginning of the end for corporate transparency? *J. Financ. Crime* 30 (2), 305–308. <https://doi.org/10.1108/JFC-03-2023-293>.
- Thorpe, C., 2021. *Implied Trusts and Beneficial Ownership in Modern UK Tax Law*. Spiramus Press, London.
- Transparency International, 2015. India: Beneficial ownership transparency. available at: https://www.transparency.org/files/content/publication/2015_BOCountryReport_India.pdf. accessed 20.11.2023.
- Turner, M., Evstropova, M., 2020, April 24. Managing risks when relying on UBO registers. available at Kroll: <https://www.kroll.com/en/insights/publications/financial-compliance-regulation/managing-risks-relying-on-ubo-registers>. accessed 19.12.2023.

- U.S. Government Accountability Office, 2019. Defense procurement: Ongoing DOD fraud risk assessment efforts should include contractor ownership. available at: <https://www.gao.gov/products/gao-20-106>. accessed 06.01.2024.
- Vaidyanathan, K.N., Mathur, A., Modak, P., 2018. A global framework for tracing Beneficial Ownership. available at: <https://t20argentina.org/wp-content/uploads/2018/06/TF9.9.8-Final.pdf>.
- Viriyasitavat, W., Da Xu, L., Bi, Z., Sapsomboon, A., 2019. New blockchain-based architecture for service interoperations in internet of things. *IEEE Trans. Comput. Soc. Syst.* 6 (4), 739–748. <https://doi.org/10.1109/TCSS.2019.2924442>.
- Wang, Q., Liu, Y., 2023. Blockchain for public safety: A survey of techniques and applications. *J. Saf. Sci. Resil.* 4 (4), 389–395. <https://doi.org/10.1016/j.jnlsr.2023.09.001>.
- Warren, J.J.D., Moffitt, K.C., Byrnes, P., 2015. How big data will change accounting. *Account. Horiz.* 29 (2), 397–407. <https://doi.org/10.2308/acch-51069>.
- Weber, R.H., 2025. From traditional to decentralized governance in the crypto environment: legal framework for decentralized autonomous organizations and decentralized finance. In: Lee, J., Darbellay, A. (Eds.), *A Research Agenda for Financial Law and Regulation*. UK Edward Elgar Publishing, Cheltenham, pp. 39–62. <https://doi.org/10.4337/9781803929996.00010>.
- Woodsome, J., Ramachandran, V., 2018. Fixing AML can new technology help address the de-risking dilemma? available at: <https://www.cgdev.org/sites/default/files/fixing-aml-can-new-technology-help-address-de-risking-dilemma.pdf>. accessed 20.11.2023.
- Xu, L.D., Viriyasitavat, W., 2019. Application of blockchain in collaborative internet-of-things services. *IEEE Trans. Comput. Soc. Syst.* 6 (6), 1295–1305. <https://doi.org/10.1109/TCSS.2019.2913165>.
- Yessenbayev, O., Nguyen, D.C.D., Jeong, T., Kang, K.J., Kim, H.R., Ko, J., Park, J.-Y., Roh, M.-S., Comuzzi, M., 2024. Combining blockchain and IoT for safe and transparent nuclear waste management: A prototype implementation. *J. Ind. Inf. Integr.* 39, 100596. <https://doi.org/10.1016/j.jii.2024.100596>.
- Yumna, H., Khan, M.M., Ikram, M., Ilyas, S., 2019. Use of blockchain in education: A systematic literature review. In: *Intelligent Information and Database Systems*. 11th Asian Conference, ACIIDS 2019. Springer International Publishing. https://doi.org/10.1007/978-3-030-14802-7_17.
- Zetzsche, D.A., Arner, D.W., Buckley, R.P., 2020. Decentralized finance. *J. Financ. Regul.* 6 (2), 172–203. <https://doi.org/10.1093/jfr/fjaa010>.

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